# Gradient descent



# Model training

**Linear regression**

Least squares loss

$$\min_{\vec{\mathbf{w}}} \sum_{(\vec{\mathbf{x}},y)\in \text{train set}} (y - \vec{\mathbf{w}} \cdot \vec{\mathbf{x}})^2$$

*How to minimize a function?*

Compute derivative, set to 0, and solve.

Quadratic polynomial:

$$ax^2 + bx + c = 0$$

Explicit solution:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Problems?

- Can't always explicitly solve (e.g. degree-5 polynomial).
- Using explicit solution can be time-consuming (Lin. reg.: $\mathcal{O}(N^{2.3})$).

Alternative: **iterative** methods.



# Iterative approach

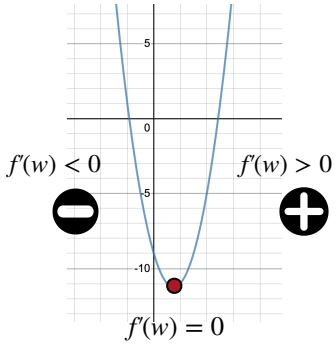
Consider $f(w) = 4w^2 - 6w - 9$.

*What do we know about the derivative's behavior, to the left of, right of, and at the minimum?*

Idea: Start somewhere on the curve and "walk down" toward its minimum.

If $f'(w) < 0$ (i.e. negative), increase $w$ (i.e. move right).

If $f'(w) > 0$ (i.e. positive), decrease $w$ (i.e. move left).








# Iterative approach

**Idea:** Start somewhere on the curve and "walk down" toward its minimum.

If $f'(w) < 0$, increase $w$ (i.e. move right).

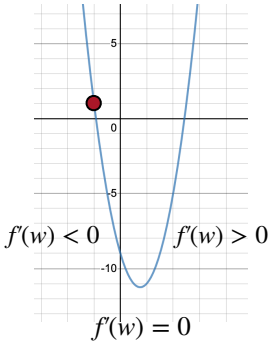
If $f'(w) > 0$, decrease $w$ (i.e. move left).

**Fill in table.** Initialize $w = -1$ and update it by $\pm 1$ according to above.

*iClicker: Write value of $w$ after 3 updates (blue cell). Any problems?*

| $w$ | $f(w)$ |
|---|---|
| -1 | 1 |
| | |
| | |
| | |

$f(w) = 4w^2 - 6w - 9$



# Iterative approach

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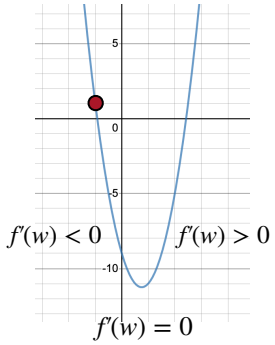
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**Fill in table.** Initialize $w = -1$ and update it by $\pm 1$ according to above.

*iClicker: Write value of $w$ after 3 updates (blue cell). Any problems?*

| $w$ | $f(w)$ |
|---|---|
| -1 | 1 |
| 0 | -9 |
| 1 | -11 |
| 0 | -9 |

$f(w) = 4w^2 - 6w - 9$






# Iterative approach

**Gradient descent update rule**

$w \leftarrow w - \eta \cdot f'(w)$, where
$\eta$ ("eta"): learning rate

**Idea:** Update $w$ based on the derivative.

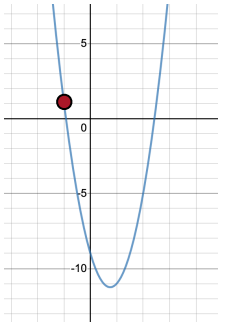
**Fill in table.** How would we update $w$?

$f'(w) = 8w - 6$
Let $\eta = 0.1$.

*Write value of $w$ after one update.*

| $w$ | $f(w)$ | $f'(w)$ |
|---|---|---|
| -1 | 1 | -14 |
| | | |
| | | |

$f(w) = 4w^2 - 6w - 9$






# Iterative approach

**Gradient descent update rule**

$w \leftarrow w - \eta \cdot f'(w)$, where
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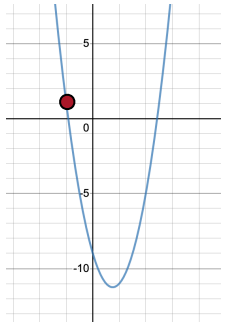
$f'(w) = 8w - 6$
Let $\eta = 0.1$.

*Write value of $w$ after one update.*

| $w$ | $f(w)$ | $f'(w)$ |
|---|---|---|
| -1 | 1 | -14 |
| **0.4** | -10.76 | -2.8 |
| 0.68 | -11.23 | -0.56 |

$w \leftarrow w - \eta \cdot f'(w)$
$w \leftarrow -1 - (0.1)(-14)$
$w \leftarrow -1 + 1.4$
$w \leftarrow 0.4$

$f(w) = 4w^2 - 6w - 9$






## Multidimensional case: Gradient

# Single- vs. multi-variable calculus

**Single-variable calculus**

**one input** → one output

$f_1(x) = x^2$

$f_2(x) = \ln(x^3)$

…

**Multi-variable calculus**

**multiple inputs** → multiple outputs

$f_3(x, y, z) = x^2y + x^3z^2$

$f_4(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}) = \begin{bmatrix} x_1 + \ln(x_2^3) \\ x_3 - x_1 \end{bmatrix}$





# Compute univariate derivative

**Single-variable calculus**

**one input** → one output

$f_1(x) = x^2$

$f_2(x) = \ln(x^3)$

…

**Try it out:** Compute the derivative of $f_2$.
*(Write answer in iClicker poll.)*





# Compute univariate derivative

**Single-variable calculus**

**one input** → one output

$f_1(x) = x^2$

$f_2(x) = \ln(x^3)$

…

**Try it out:** Compute the derivative of $f_2$.
*(Write answer in iClicker poll.)*

**Answer:**

$$\frac{df_1}{dx} = 2x$$

Use chain rule for $f_2$ (let $u = x^3$):

$$\frac{df_2}{dx} = \frac{df_2}{du}\frac{du}{dx} = \frac{1}{u} \cdot 3x^2$$
$$= \frac{3x^2}{x^3} = \frac{3}{x}$$



# Compute partial derivatives

**Multi-variable calculus**

**multiple inputs** → multiple outputs

$f_3(x, y, z) = x^2y + x^3z^2$

$f_4(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}) = \begin{bmatrix} x_1 + \ln(x_2^3) \\ x_3 - x_1 \end{bmatrix}$

**Partial derivative:** Derivative of multivar. function w.r.t. a single variable.

$$\frac{\partial f_3}{\partial x}, \frac{\partial f_3}{\partial y}, \frac{\partial f_3}{\partial z}$$

($\partial$: `\partial` in Latex)

**How to compute a partial derivative**

1. Treat all other variables as **constants**.
2. Compute the derivative w.r.t. the variable of interest.

$f_3(x,y,z) = x^2y + x^3z^2$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$



# Compute partial derivatives

**How to compute a partial derivative**

1. Treat all other variables as **constants**.
2. Compute the derivative w.r.t. the variable of interest.

**Try it out:** Compute $\frac{\partial f_3}{\partial y}$.

*(Write in iClicker poll.)*

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$





# Compute partial derivatives

**How to compute a partial derivative**

1. Treat all other variables as **constants**.
2. Compute the derivative w.r.t. the variable of interest.

**Try it out:** Compute $\frac{\partial f_3}{\partial y}$.

*(Write in iClicker poll.)*

**Answers:**

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$

$$\frac{\partial f_3}{\partial y} = x^2 + 0 = x^2$$

$$\frac{\partial f_3}{\partial z} = 0 + x^3(2z) = 2x^3z$$



# Compute gradient

**Partial derivatives example**

$$f_3(x,y,z) = x^2y + x^3z^2$$

$$\frac{\partial f_3}{\partial x} = 2xy + 3x^2z^2$$

$$\frac{\partial f_3}{\partial y} = x^2 + 0 = x^2$$

$$\frac{\partial f_3}{\partial z} = 0 + x^3(2z) = 2x^3z$$

**Gradient:** Vector containing the partial derivatives w.r.t. all input variables of a multivariate function.

$$\nabla f_3 = \begin{bmatrix} \frac{\partial f_3}{\partial x} \\ \frac{\partial f_3}{\partial y} \\ \frac{\partial f_3}{\partial z} \end{bmatrix} = \begin{bmatrix} 2xy + 3x^2z^2 \\ x^2 \\ 2x^3z \end{bmatrix}$$

($\nabla$: `\nabla` in Latex)





# Poll

Consider the function

$$f(x, y, z) = x^3zy^2 - xz + yz$$

Which of the following is the gradient of $f$, i.e. $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{bmatrix}$?

(a) $\begin{bmatrix} 3x^2y^2 \\ y \\ 3x^2y^2 - x \end{bmatrix}$ (b) $\begin{bmatrix} 3x^2zy^2 - z \\ 2x^3zy + z \\ x^3y^2 - x + y \end{bmatrix}$ (c) $\begin{bmatrix} 2x^3zy + z \\ x^3y^2 + y \\ 3x^2zy^2 - z + y \end{bmatrix}$

**Answer: (b).**